

karma

karma

2026-09-02

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
student_data <- read.csv("C:/Users/LENOVO/OneDrive/Documents/student_data.csv")
student_data
```

##	No.	Name	Gender	Age	Attendance....	Marks....	Result	
## 1	1	Aarav Patel	Male	18		92	85	Pass
## 2	2	Diya Shah	Female	17		88	78	Pass
## 3	3	Rohan Mehta	Male	19		75	62	Pass
## 4	4	Anaya Desai	Female	18		95	91	Pass
## 5	5	Kabir Joshi	Male	17		68	42	Pass
## 6	6	Meera Patel	Female	19		82	56	Pass
## 7	7	Arjun Shah	Male	18		70	35	Fail
## 8	8	Isha Mehta	Female	17		90	88	Pass
## 9	9	Vivaan Desai	Male	19		65	39	Fail
## 10	10	Siya Joshi	Female	18		85	72	Pass

```
head(student_data)
```

##	No.	Name	Gender	Age	Attendance....	Marks....	Result	
## 1	1	Aarav Patel	Male	18		92	85	Pass
## 2	2	Diya Shah	Female	17		88	78	Pass
## 3	3	Rohan Mehta	Male	19		75	62	Pass
## 4	4	Anaya Desai	Female	18		95	91	Pass
## 5	5	Kabir Joshi	Male	17		68	42	Pass
## 6	6	Meera Patel	Female	19		82	56	Pass

```
summary(student_data)
```

##	No.	Name	Gender	Age	Attendance....
## Min.	: 1.00	Length :10	Length :10	Min. :17.00	Min. :65.00
## 1st Qu.:	3.25	N.unique :10	N.unique : 2	1st Qu.:17.25	1st Qu.:71.25
## Median :	5.50	N.blank : 0	N.blank : 0	Median :18.00	Median :83.50
## Mean :	5.50	Min.nchar: 9	Min.nchar: 4	Mean :18.00	Mean :81.00

```
## 3rd Qu.: 7.75    Max.nchar:12    Max.nchar: 6    3rd Qu.:18.75    3rd Qu.:89.50
## Max.      :10.00                                Max.      :19.00    Max.      :95.00
## Marks....      Result
## Min.      :35.00    Length    :10
## 1st Qu.:45.50    N.unique  : 2
## Median   :67.00    N.blank   : 0
## Mean     :64.80    Min.nchar: 4
## 3rd Qu.:83.25    Max.nchar: 4
## Max.     :91.00
```

```
str(student_data)
```

```
## 'data.frame':    10 obs. of  7 variables:
## $ No.          : int  1 2 3 4 5 6 7 8 9 10
## $ Name         : chr  "Aarav Patel" "Diya Shah" "Rohan Mehta" "Anaya Desai" ...
## $ Gender       : chr  "Male" "Female" "Male" "Female" ...
## $ Age          : int  18 17 19 18 17 19 18 17 19 18
## $ Attendance...: int  92 88 75 95 68 82 70 90 65 85
## $ Marks....    : int  85 78 62 91 42 56 35 88 39 72
## $ Result       : chr  "Pass" "Pass" "Pass" "Pass" ...
```

```
colSums(is.na(student_data))
```

```
##           No.           Name           Gender           Age Attendance....
##           0             0             0             0             0
## Marks....      Result
##           0             0
```

```
median(student_data$Marks)
```

```
## [1] 67
```

```
var(student_data$Marks)
```

```
## [1] 446.4
```

```
sd(student_data$Marks)
```

```
## [1] 21.12818
```

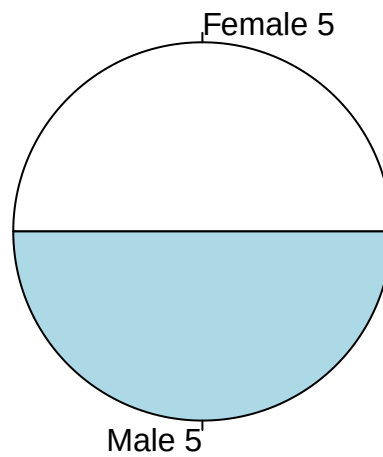
```
gender_count <- table(student_data$Gender)
```

```
gender_count
```

```
##
## Female    Male
##      5      5
```

```
pie(
  gender_count,
  main = "Gender Distribution",
  labels = paste(
    names(gender_count),
    gender_count
  )
)
```

Gender Distribution



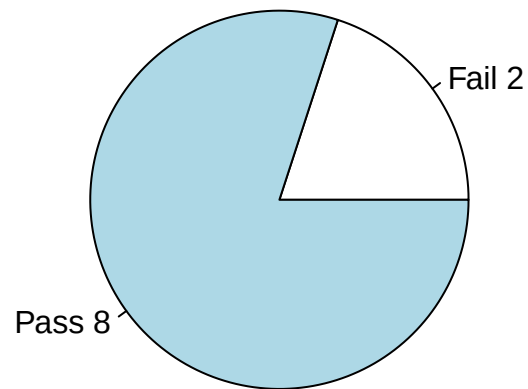
```
result_count <- table(student_data$Result)
```

```
result_count
```

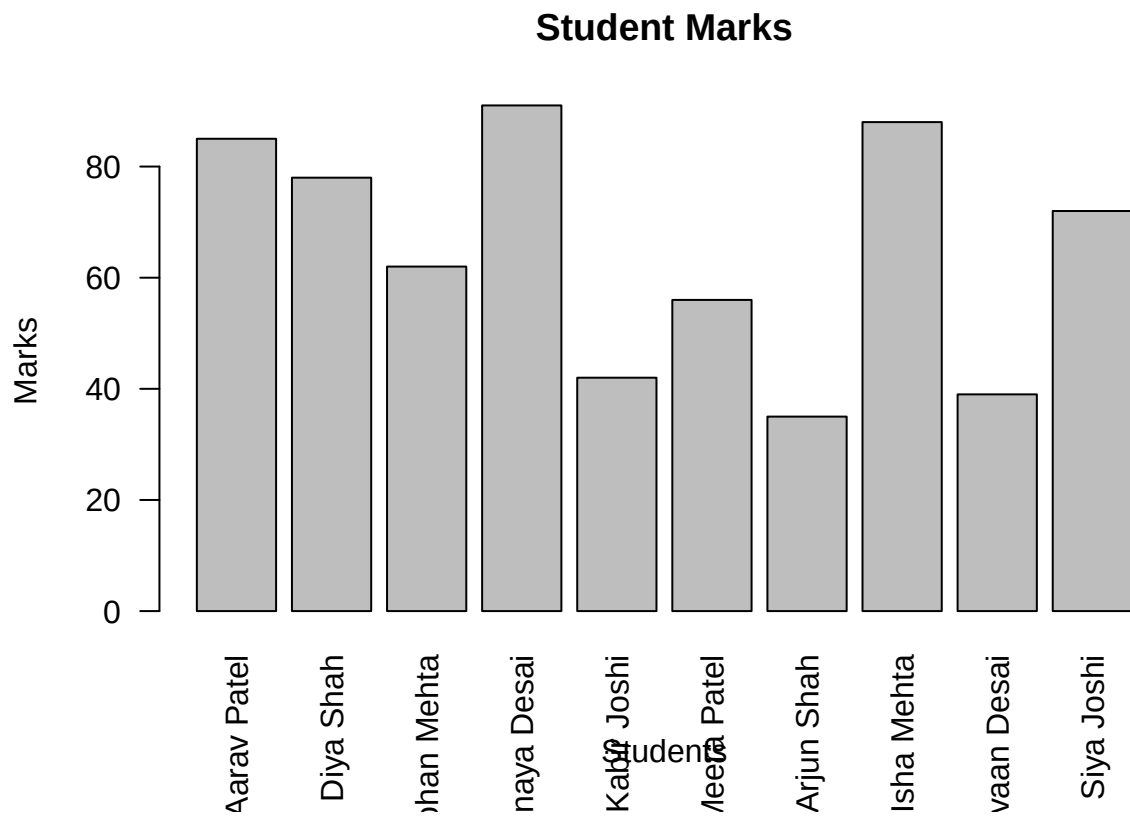
```
##
## Fail Pass
##    2    8
```

```
pie(
  result_count,
  main = "Student Result Distribution",
  labels = paste(
    names(result_count),
    result_count
  )
)
```

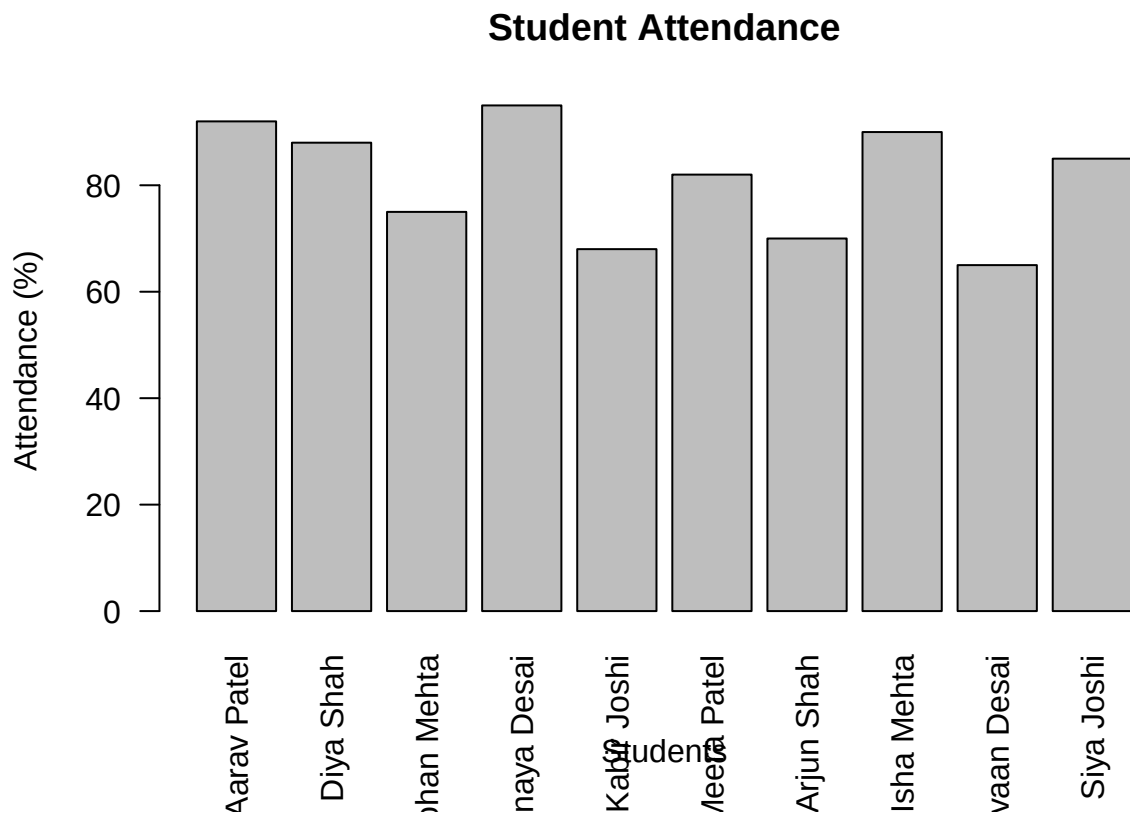
Student Result Distribution



```
barplot(  
  student_data$Marks,  
  names.arg = student_data$Name,  
  main = "Student Marks",  
  xlab = "Students",  
  ylab = "Marks",  
  las = 2  
)
```



```
barplot(
  student_data$Attendance,
  names.arg = student_data$Name,
  main = "Student Attendance",
  xlab = "Students",
  ylab = "Attendance (%)",
  las = 2
)
```

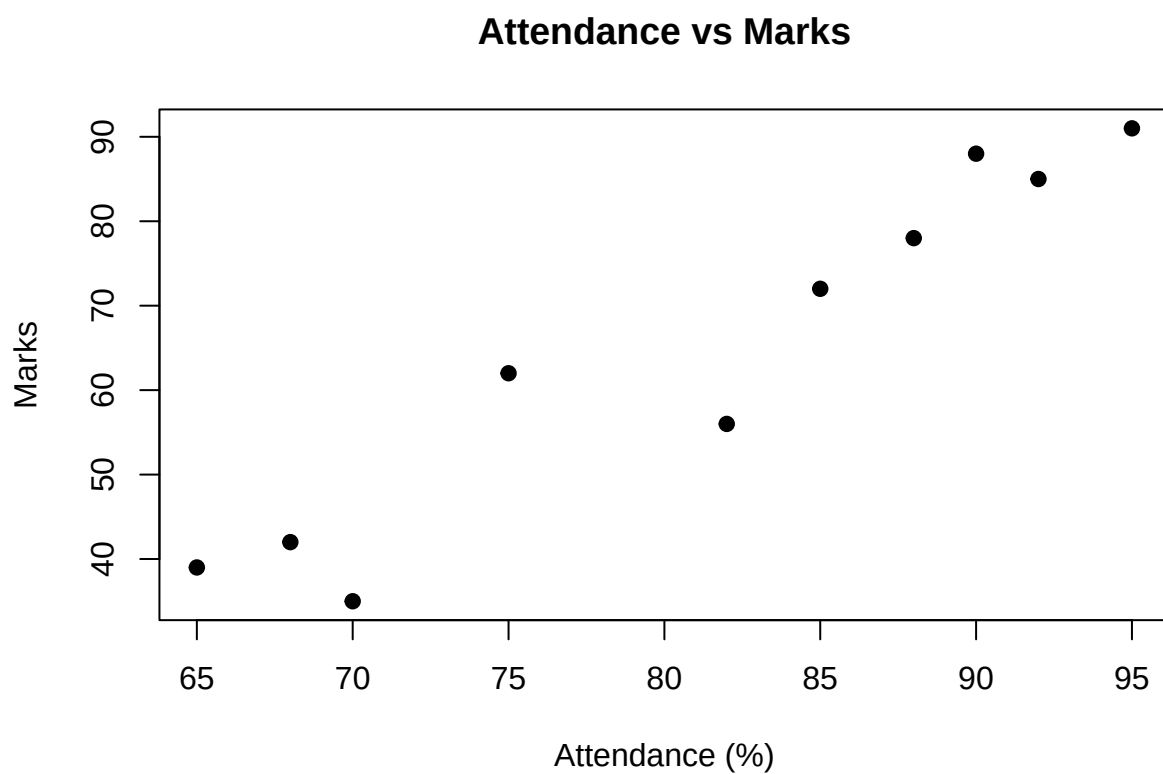


```
cor(
  student_data$Attendance,
  student_data$Marks
)
```

```
## [1] 0.9580591
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.